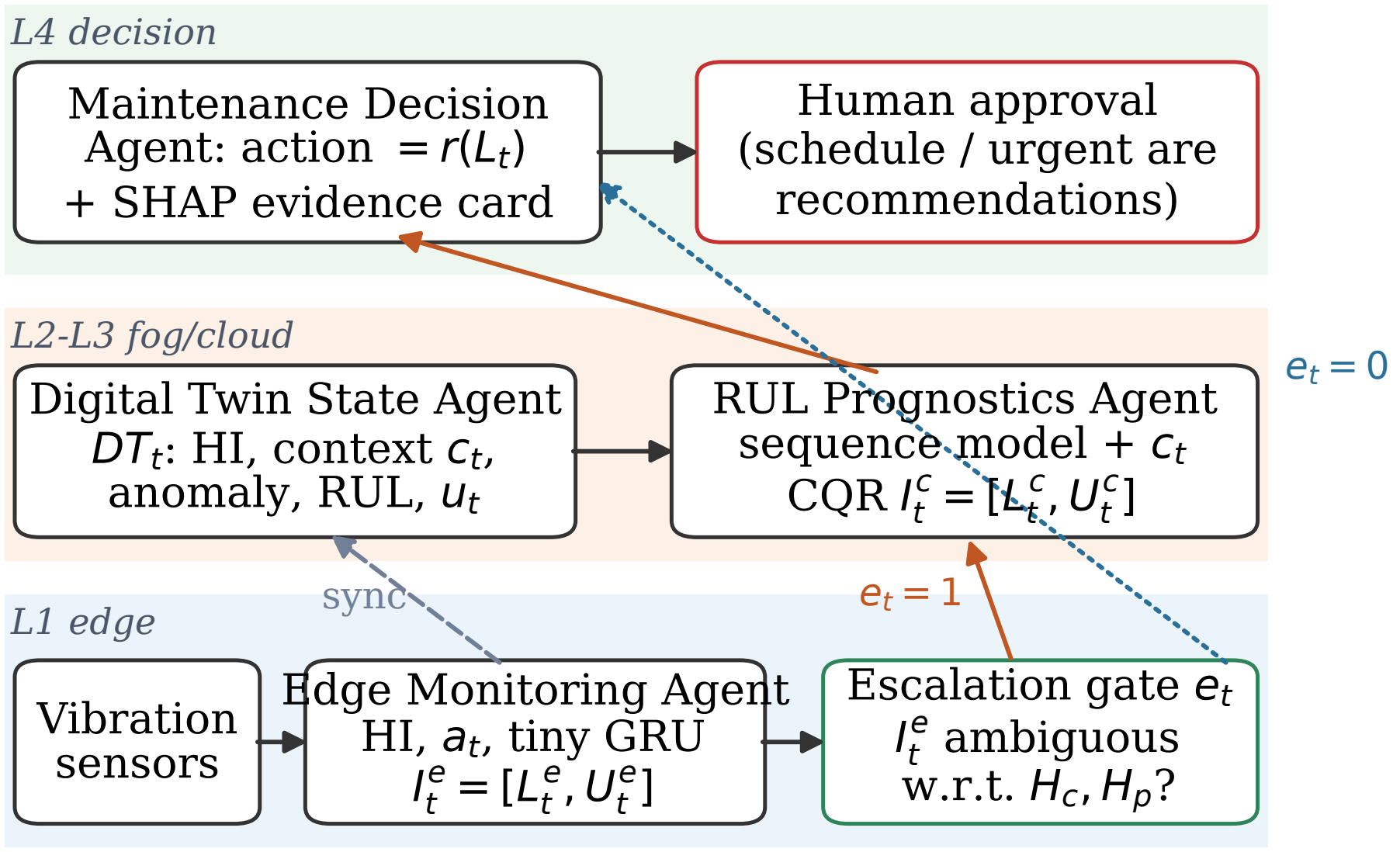

L4 decision
Maintenance Decision Agent: action $= r(L_t)$ + SHAP evidence card
Human approval (schedule / urgent are recommendations)
L2-L3 fog/cloud
Digital Twin State Agent DT_t: HI, context c_t, anomaly, RUL, u_t
RUL Prognostics Agent sequence model + c_t CQR $I_t^c = [L_t^c, U_t^c]$
$e_t = 0$
sync
$e_t = 1$
L1 edge
Vibration sensors
Edge Monitoring Agent HI, a_t, tiny GRU $I_t^e = [L_t^e, U_t^e]$
Escalation gate e_t I_t^e ambiguous w.r.t. H_c, H_p?